CLAIMS

What is claimed is:

1. A method for secure communication between a sender and a receiver, comprising the steps of:

- at the sender, generating a first secret key, K_1, which is not transmitted to the receiver;
- at the sender, encrypting a message M using K_1 to produce an encrypted message;
- transmitting the encrypted message from the sender to the receiver;
- transmitting a prefix data block P from the sender to the receiver;
- transmitting a suffix data block S from the sender to the receiver;
- at the sender, deriving a second key, K_2, from K_1 and the combination of P, the encrypted message, and S;
- transmitting K_2 from the sender to the receiver; and
- at the receiver, decrypting the combination of P, the encrypted message, and S using K_2.

2. The method of claim 1, wherein the encrypted message, the prefix P, the suffix S, and the second key K_2 are transmitted at different points in time.

3. The method of claim 1, wherein at least two of the transmissions from the group consisting of the encrypted message, the prefix P, the suffix S, and the second key K_2, are performed over different communication channels.

4. The method of claim 1, wherein the step of deriving K_2 comprises computing a cryptographic hash of the concatenation of P, the encrypted message, and S, and using the hash and K_1 as inputs to a key derivation function.

5. A system for secure communication, comprising:

- a sender device configured to:
  - generate a non-transmitted first key K_1;

  - encrypt a message M with K_1;
  - transmit the encrypted M, a prefix P, and a suffix S in separate transmissions;
  - derive a second key K_2 from K_1 and the combination of P, M, and S; and
  - transmit K_2;
- a receiver device configured to:
  - receive the encrypted M, P, S, and K_2;
  - assemble the received P, M, and S; and
  - decrypt the assembled data using K_2.

6. The system of claim 5, wherein the sender device is configured to perform the transmissions asynchronously.